

RAFAEL COSTA

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EDUCATION

Queen's University

B.A.Sc. Electrical Engineering (Innovation Stream), GPA: 4.0

Kingston, ON

Sept. 2023 – Present

- Dean's List (2023, 2024, 2025)
- Relevant Coursework: Signal Processing, Computer Architecture, Electronics, Electric Machines, Solid-State Devices

PROFESSIONAL EXPERIENCE

Geotab

Hardware Reliability & Sustaining Intern

Oakville, ON

May 2026 – Present

- Diagnosed a previously unknown MCU cold-boot bug during low-temperature reliability testing of a large production batch; root-caused via serial-log analysis, enabling the firmware team to recover all affected devices.
- Automated firmware smoke, FOTA, and long-term soak testing using Python and Raspberry Pi relay-controlled power cycling, eliminating manual test steps and gating GitLab CI releases; demoed to engineering leadership.
- Led multi-phase validation (smoke, FOTA, long-term soak) across multiple firmware releases for a telematics device, uncovering a critical regression and verifying firmware rollback-protection worked as intended.
- Designed a multi-pin pogo-pin test breakout board in Altium Designer with ESD protection; validated cellular modem firmware variants via AT commands and FOTA updates.

Queen's University, Dept. of Physics

Research Assistant – BVEX Interferometry Experiment

Kingston/Timmins, ON

Summer 2025

- Designed and implemented power distribution, heater control, and position-tracking systems (LabJack T7, relay boards, accelerometers, gyroscopes, Raspberry Pi 5 at 1 kHz), validated through thermal-vacuum testing and a field campaign.
- Programmed multithreaded C code with UDP servers to control relays, collect telemetry, and regulate system temperatures within $\pm 2^{\circ}\text{C}$.
- Collaborated with Canadian and French Space Agency engineers during a balloon launch campaign, successfully deploying electronics that operated reliably at 15 km altitude; co-authored 2 **SPIE** conference papers on the telescope and RFSoc backend.

PROJECTS

Autonomous RC Car

Control & Navigation Lead

Kingston, ON

Winter 2026

- Developed control and perception software (ZeroMQ publish-subscribe architecture) for a 4-person team building a PiCar-X autonomous taxi, implementing PID lane-keeping and Coral TPU-guided trajectory following for camera-guided turns.
- Integrated camera, grayscale, and ultrasonic sensors with GPIO-controlled turn signals and brake lights to achieve autonomous lane following, obstacle avoidance, and dynamic pickup/drop-off routing.
- Designed a mission state machine to manage high-level driving behavior, with a pivot-turn fallback mode for camera-guidance loss to improve robustness across the test course.

Smith Hyperloop – Tunnel Boring Machine

Electrical Team Manager

Kingston, ON

June 2025 – April 2026

- Leading design and testing of TBM electrical systems (power distribution, navigation controls), enabling reliable integration for an international tunneling competition.
- Coordinating subteam deliverables across mechanical, software, and electrical groups to ensure system integration under tight deadlines.

SKILLS

Hardware & EDA: Altium Designer, KiCad, LTSpice, SolidWorks, Propel PLM, Quartus II, PCB Design, Schematics
Embedded Systems: UART, SPI, I2C, USB, SWD, Cellular (AT Commands, FOTA), CAN/Telematics, Raspberry Pi
Programming & DevOps: C/C++, Python (NumPy, Pandas, Streamlit), VHDL, Bash, Git, GitLab CI/CD, Linux
Testing & Lab: Environmental Stress Screening (ESS, SAE J1455), Oscilloscopes, DMMs, Transient Generators